

31) A system of linear equations with more equations than unknowns is sometimes called an overdetermined system. Can such a system be consistent? Illustrate your answer with a specific system of three equations in two unknowns.

Soln:

$$x + 2y = 4$$

$$2x + 4y = 8$$

$$3x + 6y = 12$$

The augmented matrix is:

$$\text{Row 2} = \text{Row 2} - 2 \times \text{Row 1}$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \\ 3 & 6 & 12 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 0 \\ 3 & 6 & 12 \end{bmatrix}$$

$$\text{Row 3} = \text{Row 3} - 3 \times \text{Row 1}$$

Row 2 and Row 3 have

no pivots. Column 1 is the

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

only pivot column. $\therefore x$ is basic variable, y is free variable.

By Theorem 2, there is no row of the form $[0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$.

$\therefore$ The system is consistent. $\square$